CHAPTER 4 EVIDENCE PACKAGE: SYSTEM ANALYSIS AND DESIGN

> **What this file is.** The evidence a writer needs for Chapter 4, condensed to "what may be stated, on what basis, and where the full detail lives". It is the Chapter 4 entry point of the nine-file handover set. Repository `student-beneficiary-system`, branch `fix/sbms-completion-work`, commit `d6b5032`.
>
> **Where the full detail lives.** Prose: `CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` (§4.1–§4.8). Architecture: `SBMS_SYSTEM_ARCHITECTURE_AND_MODULE_MAP.md`. Schema: `SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md`. Interfaces: `SBMS_API_ENDPOINT_CATALOGUE.md`, `SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md`. Controls: `SBMS_SECURITY_AND_ACCESS_CONTROL_CATALOGUE.md`. Figures: `DIAGRAM_SPECIFICATIONS.md`.

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1. What Chapter 4 may assert, and on what basis

Assertion the chapter needs	Basis (code-level)	Status
Three-tier system: React SPA, Express API, PostgreSQL, Python ML service	`frontend/`, `backend/`, `ml-service/`, `docker-compose.yml`	VERIFIED AND IMPLEMENTED
Three roles, enforced server-side, role re-read per request	`backend/src/middleware/auth.js` and route guards	VERIFIED AND IMPLEMENTED
45 HTTP endpoints in ten families	`SBMS_API_ENDPOINT_CATALOGUE.md` §2–§4	VERIFIED AND IMPLEMENTED
13 tables, 14 forward migrations with a ledger and checksums	`backend/src/db/migrations/`, `backend/src/db/schema.sql`; schema doc §2–§3	VERIFIED AND IMPLEMENTED
Screening compares each new record with the whole registry, then scores pairs in the ML service	`backend/src/services/duplicateCheck.js`, `duplicateAggregation.js`, `mlDuplicateReview.js` (which spawns `ml-service/predict.py --server`)	VERIFIED AND IMPLEMENTED
The ML score is advisory: an administrator decides, with a recorded reason	`backend/src/controllers/duplicateFlags.controller.js`; `duplicate_flags` state machine	VERIFIED AND IMPLEMENTED
Payments are gated in the database as well as the API	schema doc §9 (constraints and triggers)	VERIFIED AND IMPLEMENTED
QR codes identify a student record only — never identity, eligibility or payment	`backend/src/controllers/verification.controller.js`; 10 route tests	VERIFIED AND IMPLEMENTED
Design is reproducible from specifications alone	24 figure specifications, each with implementation references	Specifications complete; drawings not produced

2. Requirement analysis: the numbers a writer must use

Group	Count	Verified	Partial	Deviated	Gaps
Functional requirements delivered (FR-01…FR-30)	30	24	6	0	0
Proposal requirements carried as deviations (FR-31…FR-33)	3	0	0	3	0
Functional gaps reported as gaps (FR-34…FR-37)	4	0	0	0	4
Non-functional requirements (NFR-01…NFR-21)	21	14	7	0	0

The six partial FRs are FR-12, FR-19, FR-25, FR-26, FR-28 and FR-29; the seven partial NFRs are NFR-01, NFR-13, NFR-14, NFR-16, NFR-18, NFR-19 and NFR-21. These are counted from the status cells in `SBMS_REQUIREMENTS_AND_SCOPE_CATALOGUE.md` §6–§8, not maintained as a separate figure.

The deviation the chapter must state, not hide (G-21): the proposal named a Random Forest over four similarity features; the delivered classifier is XGBoost over six features (`ml-service/model/`). Three of the six inputs are the proposal's features; three are added signals. Chapter Four states this in the FR-31 row of §4.2.2 and again in §4.4.4, and Chapter Six repeats it in §6.2.4.

3. Analysis artefacts required, and what each must encode

Artefact	Must encode	Specification
System context (Figure 4.1)	SBMS as one process; four external actors; three external systems	`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` §4.3.1
Use case (Figure 4.2)	Three actors outside the boundary; 14 use cases; `<<include>>` screening	§4.3.2
Activity: registration and screening (Figure 4.3)	Four swimlanes; validation failure path; advisory-score branch	§4.3.3
Activity: review, approval, payment gate (Figure 4.4)	Three swimlanes; terminal states; reminder and escalation	§4.3.4
Class diagram (Figure 4.5)	13 classes with multiplicities and state enums	§4.3.5
Sequence: registration (Figure 4.6)	Synchronous API, subprocess ML call, write-then-flag ordering	§4.3.6
Sequence: duplicate decision (Figure 4.7)	Reveal-then-decide, audit rows, terminal outcome	§4.3.7
Sequence: approval to certificate (Figure 4.8)	Snapshot, token, PDF, revocation path	§4.3.8
Sequence: public verification (Figure 4.9)	Fragment token, rate limiter, existence-only response	§4.3.9
ER diagram (Figure 4.10)	Chen notation, all 13 tables, cardinalities, FK actions	§4.3.10
ML pipeline (Figure 4.11)	Offline lane (data → features → train → artifact) and online lane (request → score → advisory flag)	§4.3.11
Architecture (Figure 4.12)	Tiers, trust boundaries, the human decision gate drawn explicitly	§4.4.1

4. Design facts that must match the code

Design area	Fact	Evidence
Deployment	Frontend and API on Azure App Service; PostgreSQL Flexible Server; ML service containerised or run as a subprocess	`SBMS_DEPLOYMENT_AND_OPERATIONS_GUIDE.md` §3–§7
Authentication	Short-lived JWT, bcrypt hashing, role claims re-validated per request	`backend/src/middleware/auth.js`
Validation	Zod schema per route, defined in each controller; five-school constraint from `backend/src/config/schools.js`; phone and identifier normalisation in `backend/src/services/studentRecordFormat.js`	`backend/src/controllers/`
Screening	Advisory lock per sweep; whole-registry comparison; `top-1` aggregation in `duplicateAggregation.js`; model version pinned per scan	`backend/src/services/`
State machines	Duplicate flag, award, disbursement and certificate lifecycles reject illegal transitions	schema doc §9
Privacy	Masked identifiers in the queue; explicit audited reveal	`backend/src/controllers/duplicateFlags.controller.js`
ML integration	Six features; 0.50 cut-off; score stored; ML failure degrades to manual review, never a silent pass	`ml-service/predict.py`; `SBMS_ML_SUBSYSTEM_DOCUMENTATION.md` §5–§6

5. Interface design: what each screen figure must show

Twelve screen figures (4.13–4.24) are specified in §4.5.1–§4.5.8 and itemised in `FIGURE_PLACEHOLDER_CHECKLIST.md` §2. Each specification names the route, the audience, the data requested, the controls that must be visible and the states the screen must render, so a screenshot can be checked against the specification rather than the specification being reverse-engineered from a screenshot.

6. What Chapter 4 must not claim

- A number of users, records or deployments: no production usage exists, so no usage figure is supportable.
- That the ML model decides duplicates: the human gate is a design control and must be drawn as a boundary.
- That figures exist: they are specified only, and every placeholder says so.
- That a phone camera scanned a printed code: no such capture exists (G-11).

7. Open items affecting Chapter 4

Item	Effect	Owner
G-03/G-04/G-05 figures and screenshots	The interface section cannot be illustrated	Author
A-2 use-case table granularity	§4.3.2 may need more or fewer use-case tables	Author
A-3 screenshot provenance	Captions must say whether the instance is Azure or local	Author

`ACADEMIC_COMPLETENESS_AND_GAP_AUDIT.md` §3 carries the full gap list; `MASTER_EVIDENCE_REGISTER.md` carries the evidence IDs behind every row above.

8. Related files

`CHAPTER_4_SYSTEM_ANALYSIS_AND_DESIGN.md` (the chapter), `DIAGRAM_SPECIFICATIONS.md` (figure register), `SBMS_DATABASE_SCHEMA_AND_DATA_DICTIONARY.md` (schema), `SBMS_UI_SCREEN_AND_WORKFLOW_INVENTORY.md` (screens).